



# Intelligence

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# Mapping your intelligence journey!

- Plot key moment when different intelligence were important for you
- Quick theory reference:
  - Spearman: Single g factor
  - Thurstone: verbal comprehension, word fluency, numerical ability, spatial visualisation, associative memory, Perceptual speed, Reasoning
  - Gardner: Linguistic Intelligence, Logical-mathematical Intelligence, Spatial Intelligence, Bodily Kinesthetics Intelligence (Body smart), Musical Intelligence, Interpersonal intelligence, Intrapersonal intelligence (Self smart), Naturalistic intelligence, Existential intelligence (possible 9th )
  - Cattell's theory of intelligence: Crystallized intelligence, Fluid intelligence
  - Sternberg: Analytical, Creative, Practical
  - Creativity: Convergent and divergent

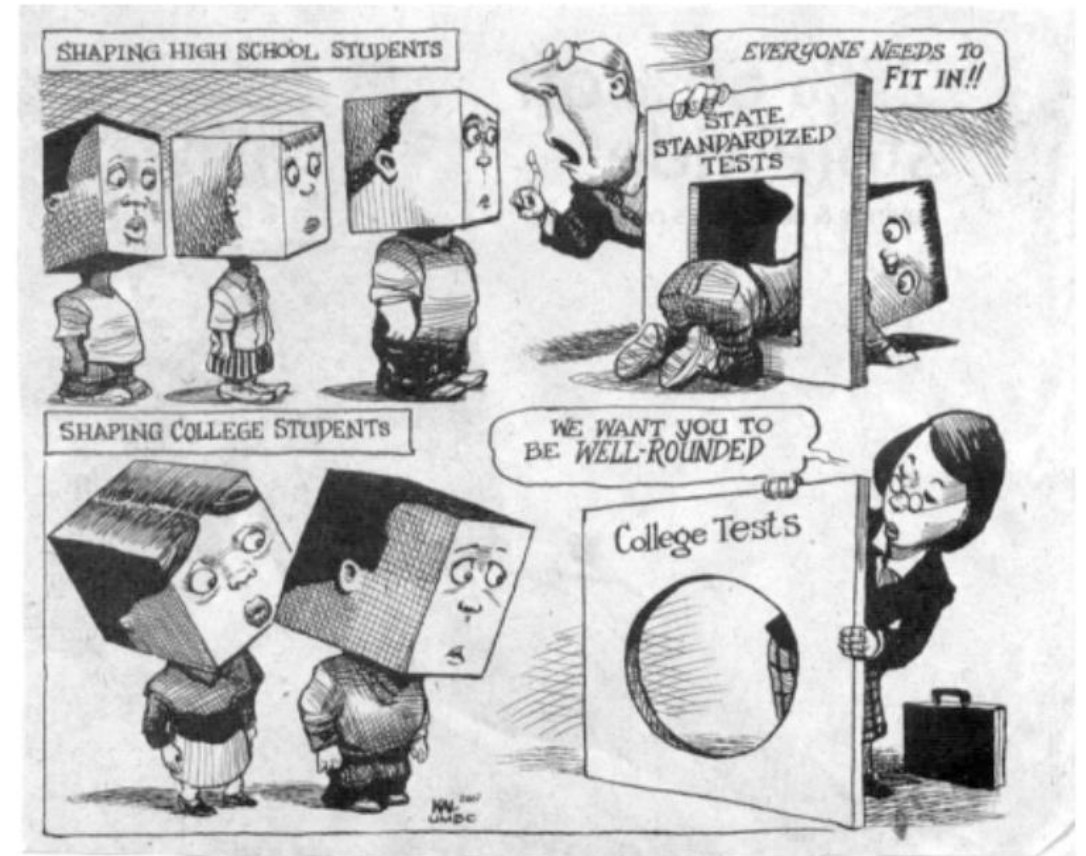


# History of testing



# Types of Cognitive ability testing

- Ability tests (Present):
  - Assess a person's innate or general cognitive capacity
  - Natural capability in specific skills
  - For example: Wechsler Adult Intelligence Scale (WAIS)
- Aptitude tests (Future):
  - Measures the potential and capacity to learn new skills
  - Persons ability to acquire knowledge or skills that have not been learned
  - For example: DAT
- Achievement tests (Past):
  - Evaluate what individuals have already learned or accomplished
  - Knowledge and skill gained through instructions or experience
  - For example: College exams



Speed test  
(time limit)

Power test  
(increasing  
difficulty)

# **Discussion: GATE as an Aptitude test**

# Insights!

- Does this test reflect aptitude required today to pursue career?
- Cultural Bias
- Narrow focus
- Coaching effect
- Learning potential in future?
- Relevance today!

MAYBE STANDARDIZED TESTING  
HAS GONE A LITTLE TOO FAR.

IS SHE  
PROFICIENT?

NEEDS  
IMPROVE-  
MENT.

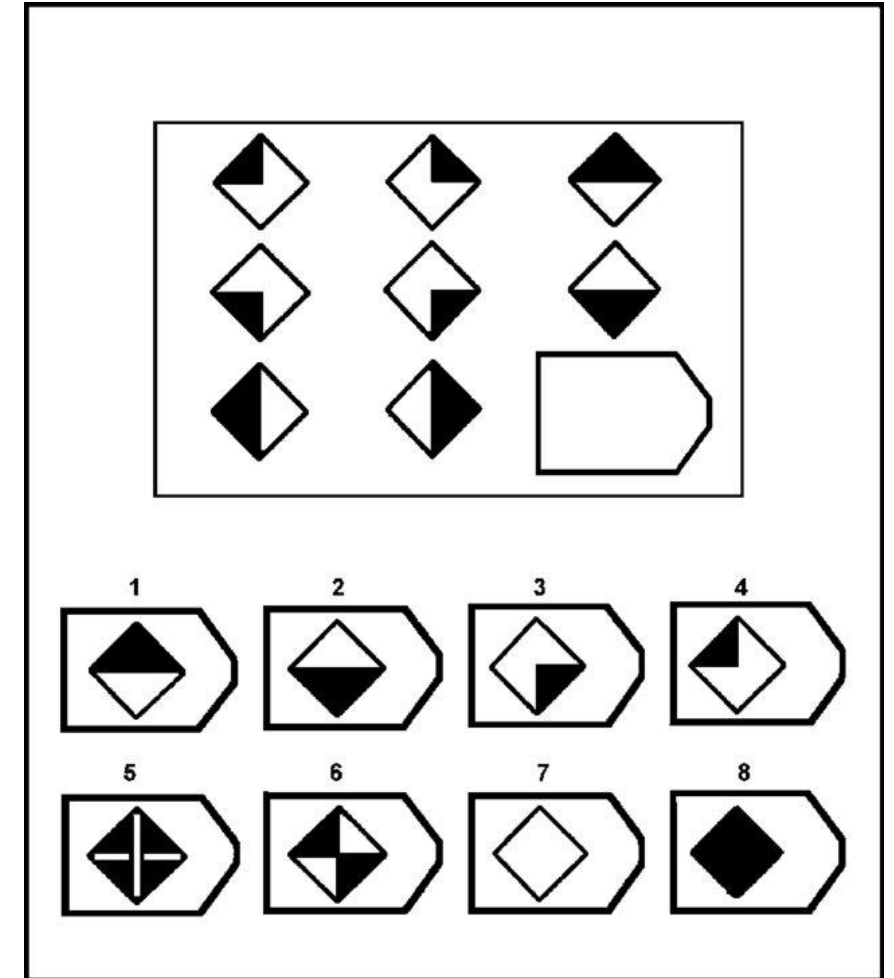
REPEAT

## Culture- fair test!



# RAVEN'S PROGRESSIVE MATRICES

- Developed by John Raven (1938)
- Designed to understand non-verbal intelligence
- Key feature:
  - Culture-fair test
  - Power test: items increase in difficulty
  - Some may include time limit but focus is on quality of answer and ability to solve complex patterns



# THOUGHTS?

- Any intellectual ability required?
- Is it culture-fair?
- Does it study ability?

# CONCLUSION

- History of testing
- Current intelligence testing
  - Ability, aptitude, achievement testing
- Culture-fair test
  - Raven's progressive matrices